Vikas Babu

Embedded Firmware Engineer

RESUME_1061

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Total Experience: 9 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 9 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: CAN, FreeRTOS, UART, JTAG, Embedded Linux, C, ARM Cortex-M, I2C, STM32, Firmware Debugging, C++, SPI

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Lead Embedded Firmware Engineer | HelioNex Labs | Jan 2023 - Present

- Validated 264 hardware test vectors across protocol and timing corner cases
- Lowered field-reset incidents by 22% with watchdog recovery and diagnostics
- Improved battery runtime by 18% with peripheral and sleep-state optimization
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Embedded Firmware Engineer | DeltaPoint Innovations | Jan 2020 - Dec 2022

- Improved battery runtime by 41% with peripheral and sleep-state optimization
- Reduced embedded task jitter by 14% using interrupt priority and queue tuning

Associate Embedded Firmware Engineer | WillowGrid Labs | Jul 2017 - Dec 2019

- Validated 397 hardware test vectors across protocol and timing corner cases
- Lowered field-reset incidents by 28% with watchdog recovery and diagnostics

PROJECTS

Motor Control Interface | JTAG, C++, C, UART

- Implemented CAN message handling, fault logging and deterministic state transitions.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Low-Power Metering Device | FreeRTOS, SPI, I2C, CAN

- Optimized interrupt-driven acquisition and sleep cycles.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Silverlake School of Engineering, 2017
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

Embedded C Standards (2022)